Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 44

Growth of Functions

Covers Chartrand & Zhang, Section 6.2

Status of this set

Ungraded practice. Solutions released with it. Every big-O claim needs its witnesses.

Part I: How to Practice This Material

Proving $f = O(g)$ is exhibiting a witness

Name a constant C and an integer k, then verify $f(n) \le C\,g(n)$ for all $n \ge k$. That is the whole proof — an existence claim, as in Lecture 26.
A useful habit: bound each term of f separately by a multiple of g, then add the constants. For $5n + 7$ against n^2: $5n \le 5n^2$ and $7 \le 7n^2$ for $n \ge 1$, so $C = 12$ works.

Disproving $f = O(g)$

Show the ratio $f(n)/g(n)$ grows without bound. If $f(n) \le C\,g(n)$ held for all large n, that ratio would stay below the fixed number C.

Three mistakes to expect

1. **Asserting $f = O(g)$ with no C and k.** The definition asks for them.
2. **Treating big-O as symmetric.** $f = O(g)$ does not give $g = O(f)$.
3. **Proving only one inequality for big-theta.** It needs both.

Part II: Practice Problems

Section A — Big-O with witnesses

1. Prove each, giving explicit C and k.

(a) $3n + 5 = O(n)$

(b) $n^2 + 3n + 2 = O(n^2)$

(c) $4n^3 + 2n = O(n^3)$

2. Show that $n^2 \neq O(n)$.

Section B — Which are $O(n)$?

3. For each function, decide whether $f(n) = O(n)$, with a reason.

(a) $f(n) = 100$

(b) $f(n) = 2n + 5$

(c) $f(n) = \lfloor n/2 \rfloor$

(d) $f(n) = n^2 + 3n + 2$

Section C — Big-theta

4. Show that $f(n) = 3n^2 + 2n$ and $g(n) = n^2$ satisfy $f = \Theta(g)$.

5. For each f, give the smallest nonnegative integer k with $f(n) = O(n^k)$.

(a) $f(n) = n^2 + n \log n$

(b) $f(n) = \dfrac{2n^3 + n + 1}{n^2 + 1}$

(c) $f(n) = (n^2 + 1)(n^3 + 1)$

Section D — Writing

6. In four to six complete sentences, explain why $f = O(g)$ does not imply $g = O(f)$, and what additional condition makes the two equivalent.